

INTCORT: TRAINING-FREE SPATIAL REASONING ENHANCEMENT FOR VISION-LANGUAGE MODELS VIA INPUT TRANSFORMATIONS AND CONFIDENCE ROUTING

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ABSTRACT

Vision-Language Models (VLMs) have demonstrated remarkable capabilities in multimodal tasks, yet they still exhibit poor ability in spatial reasoning. Existing training-dependent and training-free enhancement methods suffer from high computational costs with catastrophic forgetting and internal mechanism interference that compromises general capabilities, respectively. In this work, we first verify two key hypotheses: appropriate geometric image transformation and query-reversal transformation can recover incorrect spatial predictions, and correct predictions exhibit higher relation-token confidence than incorrect ones. Based on these findings, we propose **INTCORT**, a training-free spatial reasoning enhancement framework that constructs multiple inference views through input transformations and aggregates their predictions via relation-token confidence routing, without modifying the VLM’s internal mechanisms. Experimental results on several commonly-used benchmarks demonstrate that **INTCORT** substantially improves spatial reasoning accuracy across diverse VLMs, achieving an average improvement of 10.01% over all models and benchmarks. Compared with prior works, **INTCORT** achieves superior performance with improvements of up to 25.01%.

Index Terms— Vision-Language Models, Spatial Reasoning, Input Transformations

1. INTRODUCTION

Vision-Language Models (VLMs) have achieved remarkable progress in image captioning, visual question answering, and multimodal dialogue [1, 2, 3]. However, despite their strong general visual understanding capabilities, VLMs still struggle with poor ability in spatial reasoning, such as basic spatial relation queries like “left/right” or “above/below” [4, 5, 6]. Understanding spatial relations among objects is essential for embodied intelligence, robot navigation, and scene understanding, making this limitation a key obstacle to real-world deployment [7].

To enhance the spatial reasoning capabilities of VLMs, existing methods can be categorized into training-dependent and training-free approaches. For training-dependent methods, they typically use large-scale spatial reasoning data to fine-tune VLMs. For training-free methods, they typically employ internal attention intervention or decoding strategies such as AdaptVis [8], DoLa [9], and VCD [10]. The training-dependent methods incur

high computational costs and may cause catastrophic forgetting [11]. The training-free methods, as they intervene in the internal mechanisms of VLMs via attention intervention and decoding strategies, may degrade the VLMs’ original performance on other reasoning tasks. Therefore, these limitations motivate us to explore a training-free inference strategy that enhances spatial reasoning without modifying the VLM’s internal mechanisms, thus avoiding compromising its original capabilities.

In this work, our key intuition is that an incorrect prediction from the original input does not necessarily indicate a lack of spatial reasoning ability, as VLM predictions may depend on the input view. Geometric image transformations and query reversal provide alternative inference views from which incorrect direct predictions may be recovered. Based on this intuition, we formulate and validate two hypotheses: appropriate transformations can recover incorrect spatial relation predictions, and correct predictions tend to exhibit higher relation-token confidence than incorrect ones. Motivated by these findings, we propose **INTCORT**, a two-stage training-free framework based on input transformations and confidence routing. Specifically, **INTCORT** first uses a lightweight language model to identify spatial queries and avoid unnecessary transformations for non-spatial inputs. For spatial queries, it then constructs multiple transformed inference branches, maps the predictions produced by the VLMs back to the original answer space, and aggregates them using relation-token confidence. This multi-view evidence aggregation improves spatial reasoning reliability without modifying the VLM’s internal mechanisms.

To evaluate the effectiveness of **INTCORT**, we conduct extensive experiments on seven spatial reasoning benchmarks[5, 8, 4, 12, 13] across six mainstream VLMs, demonstrating consistent improvements over corresponding base models and existing training-free methods. Our main contributions are summarized as follows: 1) We reveal that spatial reasoning failures in VLMs can often be recovered through alternative inference views and propose **INTCORT**, a training-free spatial reasoning enhancement framework that operates exclusively at the input side via multi-view transformations and confidence routing without modifying the VLM’s internal mechanisms. 2) We verify that geometric transformations can recover incorrect spatial predictions and that correct transformed predictions tend to have higher relation-token confidence scores than incorrect direct predictions. 3) We validate the superiority and complementarity of **INTCORT** across seven commonly-used spatial reasoning benchmarks and six mainstream VLMs, without degrading performance on general

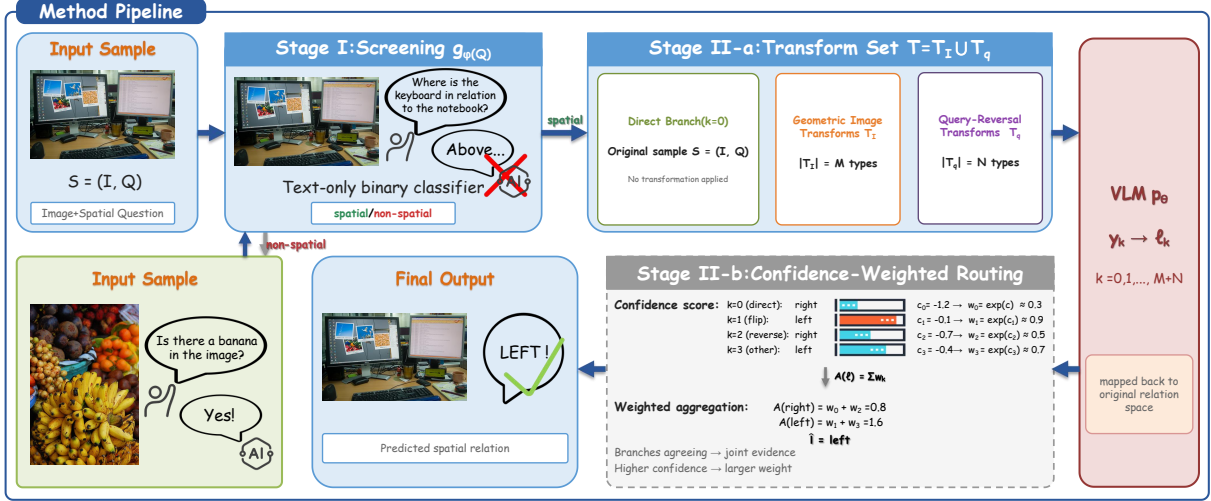


Fig. 1. Overview of the proposed INTCORT framework.

benchmarks.

2. PROPOSED METHOD

Spatial reasoning requires VLMs to understand the spatial arrangement of entities in a visual scene and express the corresponding relations in language. Formally, given an image I and a spatial reasoning question q , a VLM parameterized by θ generates a response y according to $y \sim p_{\theta}(\cdot | I, q)$. Despite their strong visual understanding capabilities, current VLMs remain unreliable in spatial reasoning. In this work, we focus on two representative types of spatial reasoning, namely the position of an entity with respect to the image and the spatial relation between two entities in the image.

2.1. Recovering Spatial Predictions via Transformations

Motivation. Intuitively, directly answering a spatial reasoning question from the original input may lead to an incorrect relation prediction. Geometric image transformations and query reversal provide complementary inference views of the same underlying spatial relation. These views may help the VLM better perceive and compare spatial cues, allowing it to produce the correct relation when direct inference fails. Based on this intuition, we formulate two hypotheses. First, appropriate transformations can recover incorrect spatial predictions. Second, correct predictions are expected to exhibit higher confidence on the generated relation token than incorrect ones. To validate these hypotheses, we conduct experiments with Qwen2-VL-7B on the Controlled-B and CV-Bench-R benchmarks. The detailed experimental setup is provided in Section 3.

Specifically, we define a geometric image transformation set $\mathcal{T}_I$ with $|\mathcal{T}_I| = M$, including transformations such as horizontal and vertical flips. We further define a query-reversal transformation set $\mathcal{T}_q$ with $|\mathcal{T}_q| = N$, where each transformation reverses the query direction by swapping the target and reference entities. For example, “Where is the mug in relation to the knife?” is transformed into “Where is the knife in relation to the mug?”, with $N = 1$ for this example. Combining the two sets, we define the complete transformation set as $\mathcal{T} = \mathcal{T}_I \cup \mathcal{T}_q$, with $|\mathcal{T}| = M + N$.

Model	Benchmark	n_{err}	n_{rec}	n_{conf}	ρ_{rec}	ρ_{conf}
Qwen2-VL-7B	Controlled-B + CV-Bench-R	141	130	107	92.20%	82.31%

Table 1. Analysis of recoverable spatial predictions and confidence.

For each sample $S = (I, Q)$ in the Controlled-B and CV-Bench-R benchmarks, we first perform direct inference using the VLM and collect the incorrectly predicted samples into $\mathcal{S}_{\text{err}} = \{S^{(i)}\}_{i=1}^{n_{\text{err}}}$, where $|\mathcal{S}_{\text{err}}| = n_{\text{err}}$ and n_{err} denotes the number of samples incorrectly predicted by direct inference. For each erroneous sample $S^{(i)} \in \mathcal{S}_{\text{err}}$, we apply every transformation $t_k \in \mathcal{T}$ to obtain $S_k^{(i)} = t_k(S^{(i)})$, where $k = 1, \dots, M + N$ and $S_k^{(i)}$ denotes the sample generated by applying the k -th transformation to $S^{(i)}$. We then define the transformed sample set corresponding to $S^{(i)}$ as $\mathcal{S}_{\text{trans}}^{(i)} = \{S_k^{(i)}\}_{k=1}^{M+N}$, with $|\mathcal{S}_{\text{trans}}^{(i)}| = M + N$.

Next, we perform VLM inference on every transformed sample in $\mathcal{S}_{\text{trans}}^{(i)}$. An erroneous sample $S^{(i)}$ is considered recovered if at least one of its $M + N$ transformed samples produces the correct relation prediction after relation alignment. We denote the total number of recovered samples by n_{rec} and define the recovery ratio as $\rho_{\text{rec}} = n_{\text{rec}}/n_{\text{err}}$. For each recovered sample, we further record the confidence score $c_{\text{rec}}^{(i)}$ of the correct transformed prediction and compare it with the confidence score $c_{\text{dir}}^{(i)}$ of the incorrect direct prediction. We denote by n_{conf} the number of recovered samples satisfying $c_{\text{rec}}^{(i)} > c_{\text{dir}}^{(i)}$, and define the confidence superiority ratio as $\rho_{\text{conf}} = n_{\text{conf}}/n_{\text{rec}}$.

As shown in Table 1, the recovery ratio ρ_{rec} reaches 92.20%, indicating that, for most samples incorrectly answered by direct inference, at least one transformed view can recover the correct spatial relation. Moreover, the confidence superiority ratio ρ_{conf} reaches 82.31%, showing that the correct transformed prediction has a higher relation-token confidence than the incorrect direct prediction in the majority of recovered samples. These results provide support for our two hypotheses and motivate the use of transformed inference views together with relation-token confi-

dence for spatial reasoning.

2.2. INTCORT Framework

Based on the two verified hypotheses, we propose INTCORT, whose overall framework is illustrated in Fig. 1. Given an input sample $S = (I, Q)$, INTCORT performs inference in two stages. The first stage determines whether the input question requires spatial reasoning, while the second stage constructs transformed inference views and aggregates their relation predictions based on relation-token confidence.

Stage I: Spatial Query Screening. This stage determines whether the input question requires spatial reasoning, as applying spatial transformations to non-spatial questions may introduce irrelevant variations. Specifically, we feed the input question Q into a lightweight language model g_ϕ , which performs binary classification such that $g_\phi(Q) \in \{\text{spatial}, \text{non-spatial}\}$. If $g_\phi(Q) = \text{non-spatial}$, we directly query the VLM using the original input; otherwise, the sample is passed to the second stage of INTCORT. This screening step prevents unnecessary transformations on non-spatial tasks and preserves the original capability of VLMs.

Stage II: Transformation and Confidence Routing. For a spatial reasoning question, we first perform direct inference on the original input and construct a set of transformed inputs using the transformation set $\mathcal{T}$ defined in Section 2.1. We denote the direct inference branch by $k = 0$ and the $M + N$ transformed branches by $k = 1, \dots, M + N$. Each branch is independently processed by the VLM to generate a response y_k , from which we extract the predicted spatial relation label ℓ_k . We then map ℓ_k back to the spatial relation space of the original input and denote the mapped relation label by ℓ_k^* , with $\ell_0^* = \ell_0$ for the direct branch. We use the confidence of the generated relation tokens to measure the reliability of each branch prediction. Let $\mathcal{P}_k$ denote the set of token positions in y_k corresponding to the tokens that constitute ℓ_k . We define the relation-token confidence score as

$$c_k = \min_{t \in \mathcal{P}_k} \log p_\theta(y_{k,t} | y_{k,<t}, I_k, Q_k), \quad (1)$$

where $y_{k,t}$ denotes the t -th generated token. For a relation label ℓ_k consisting of multiple tokens, the minimum token log-probability is used as its confidence score. We further convert c_k into a positive aggregation weight as $w_k = \exp(c_k)$. For each candidate relation label ℓ , we aggregate the weights of all branches whose mapped predictions correspond to ℓ as

$$A(\ell) = \sum_{k=0}^{M+N} \mathbb{I}[\ell_k^* = \ell] w_k, \quad (2)$$

where $\mathbb{I}[\cdot]$ denotes the indicator function. The final relation prediction is selected as $\hat{\ell} = \arg \max_{\ell} A(\ell)$. This confidence-weighted aggregation favors relation predictions with broader branch support and higher relation-token confidence.

3. EXPERIMENTS

We evaluate INTCORT on seven widely used spatial reasoning benchmarks using Qwen-VL-Chat [3], Qwen2-VL-7B [14], Qwen2.5-VL-7B [15], Qwen3-VL-8B [16], Molmo-7B [17], and LLaVA-NeXT-7B [18], and compare it with state-of-the-art

training-free methods, including AdaptVis [8], DoLa [9], and VCD [10]. We use accuracy as the evaluation metric. For the spatial query classifier g_ϕ , we use Qwen2.5-1.5B-Instruct. The geometric image transformation set is fixed as $\mathcal{T}_I = \{H, V\}$, where H and V denote horizontal and vertical flips, respectively. All experiments are conducted using NVIDIA A800 GPUs.

3.1. Benchmarks

The benchmark suite covers both controlled and real-world scenes. **Controlled-A** and **Controlled-B** are the two controlled benchmarks introduced in AdaptVis [8]. They contain clean-background images with two objects. Controlled-A consists of one large object and one small object, whereas Controlled-B contains two small objects. For real-world scenes, we use **COCO-one** and **COCO-two** from WhatsUp [4], which are constructed using images from MS COCO [13]. COCO-one evaluates the position of a single entity relative to the image, while COCO-two evaluates the spatial relation between two entities. Similarly, **VG-one** and **VG-two** are constructed from Visual Genome [12] and evaluate single-entity and two-entity spatial reasoning, respectively. Finally, we use **CV-Bench-R**, the relation subset of CV-Bench [5], to evaluate pairwise spatial reasoning in diverse real-world scenes.

3.2. Experimental Results

As shown in Table 2, INTCORT achieves the best or tied-best performance on all benchmarks with Qwen-VL-Chat, Qwen2-VL-7B, Qwen2.5-VL-7B, Qwen3-VL-8B, and Molmo-7B. Compared with the five above original model and AdaptVis (the best among all compared methods), INTCORT improves accuracy by 10.01% and 6.47%, respectively, averaged across the five models and all seven benchmarks. These results demonstrate that INTCORT achieves significant improvements in spatial reasoning across diverse vision-language models and benchmarks. When LLaVA-NeXT-7B is equipped with INTCORT, it outperforms AdaptVis on most benchmarks and achieves an average accuracy improvement of 10.52% over the original model across all benchmarks. A possible reason why INTCORT slightly underperforms AdaptVis on a few benchmarks is that it relies on the model’s inherent spatial reasoning ability and exploits it through transformed views. Since LLaVA-NeXT encodes relatively weak spatial evidence [19], INTCORT may be less effective, whereas AdaptVis directly adjusts the attention distribution and is less dependent on the model’s original spatial reasoning capability. Moreover, as shown in Table 5, applying INTCORT to Qwen2-VL-7B preserves its accuracy on the non-spatial POPE [20] and MME[21] benchmarks, indicating that our method does not compromise the model’s original capabilities on non-spatial tasks.

3.3. Complementarity Analysis

To further demonstrate the benefits of INTCORT, we investigate its complementarity with AdaptVis on LLaVA-NeXT-7B across seven spatial reasoning benchmarks. The two methods are theoretically compatible, as INTCORT enhances spatial reasoning through input transformations without modifying the VLM’s internal mechanisms, while AdaptVis improves spatial reasoning by adjusting the model’s internal attention distribution. The results are presented in Table 3. Specifically, INTCORT combined

Model	Method	VG-one	VG-two	COCO-one	COCO-two	CV-Bench-R	Controlled-A	Controlled-B
Qwen-VL Chat	Original model	41.81	36.05	44.72	57.10	54.42	64.16	63.11
	+AdaptVis	51.40 $\uparrow$ 9.59	46.35 $\uparrow$ 10.30	52.78 $\uparrow$ 8.06	57.39 $\uparrow$ 0.29	55.01 $\uparrow$ 0.59	64.97 $\uparrow$ 0.81	63.41 $\uparrow$ 0.30
	+DoLa	41.49 $\downarrow$ 0.32	37.34 $\uparrow$ 1.29	45.78 $\uparrow$ 1.06	57.36 $\uparrow$ 0.26	54.61 $\uparrow$ 0.19	64.37 $\uparrow$ 0.21	62.80 $\downarrow$ 0.31
	+VCD	39.96 $\downarrow$ 1.85	38.20 $\uparrow$ 2.15	47.06 $\uparrow$ 2.34	56.42 $\downarrow$ 0.68	55.19 $\uparrow$ 0.77	63.55 $\downarrow$ 0.61	62.28 $\downarrow$ 0.83
	+INTCORT	56.68$\uparrow$14.87	57.94$\uparrow$21.89	53.00$\uparrow$8.28	64.49$\uparrow$7.39	57.69$\uparrow$3.27	65.12$\uparrow$0.96	64.16$\uparrow$1.05
Qwen2-VL 7B	Original model	73.92	57.51	69.35	75.57	78.27	97.89	91.46
	+AdaptVis	74.14 $\uparrow$ 0.22	67.09 $\uparrow$ 9.58	70.44 $\uparrow$ 1.09	77.55 $\uparrow$ 1.98	78.85 $\uparrow$ 0.58	97.59 $\downarrow$ 0.30	91.81 $\uparrow$ 0.35
	+DoLa	79.09 $\uparrow$ 5.17	66.18 $\uparrow$ 8.67	69.46 $\uparrow$ 0.11	76.42 $\uparrow$ 0.85	78.46 $\uparrow$ 0.19	97.48 $\downarrow$ 0.41	92.22 $\uparrow$ 0.76
	+VCD	76.51 $\uparrow$ 2.59	58.00 $\uparrow$ 0.49	66.24 $\downarrow$ 3.11	74.43 $\downarrow$ 1.14	79.42 $\uparrow$ 1.15	96.69 $\downarrow$ 1.20	91.16 $\downarrow$ 0.30
	+INTCORT	79.20$\uparrow$5.28	68.67$\uparrow$11.16	70.91$\uparrow$1.56	79.26$\uparrow$3.69	82.89$\uparrow$4.62	99.10$\uparrow$1.21	99.09$\uparrow$7.63
Qwen2.5-VL 7B	Original model	61.85	58.37	68.74	77.84	88.46	95.48	96.04
	+AdaptVis	62.61 $\uparrow$ 0.76	59.66 $\uparrow$ 1.29	69.69 $\uparrow$ 0.95	78.69 $\uparrow$ 0.85	89.23 $\uparrow$ 0.77	95.48	96.95 $\uparrow$ 0.91
	+DoLa	62.18 $\uparrow$ 0.33	59.66 $\uparrow$ 1.29	67.90 $\downarrow$ 0.84	76.45 $\downarrow$ 1.39	89.38 $\uparrow$ 0.92	86.75 $\downarrow$ 8.73	95.20 $\downarrow$ 0.84
	+VCD	62.59 $\uparrow$ 0.74	59.02 $\uparrow$ 0.65	67.41 $\downarrow$ 1.33	78.69 $\uparrow$ 0.85	89.04 $\uparrow$ 0.58	91.57 $\downarrow$ 3.91	96.70 $\uparrow$ 0.66
	+INTCORT	73.81$\uparrow$11.96	72.96$\uparrow$14.59	70.80$\uparrow$2.06	81.82$\uparrow$3.98	90.77$\uparrow$2.31	96.08$\uparrow$0.60	99.39$\uparrow$3.35
Qwen3-VL 8B	Original model	69.94	76.39	72.36	81.25	91.15	99.40	97.26
	+AdaptVis	72.21 $\uparrow$ 2.27	77.67 $\uparrow$ 1.28	71.69 $\downarrow$ 0.67	81.53 $\uparrow$ 0.28	91.15	99.40	97.48 $\uparrow$ 0.22
	+DoLa	70.61 $\uparrow$ 0.67	77.54 $\uparrow$ 1.15	72.97$\uparrow$0.61	80.11 $\downarrow$ 1.14	91.15	98.49 $\downarrow$ 0.91	97.56 $\uparrow$ 0.30
	+VCD	71.18 $\uparrow$ 1.24	77.39 $\uparrow$ 1.00	70.52 $\downarrow$ 1.84	81.25	91.35 $\uparrow$ 0.20	97.59 $\downarrow$ 1.81	97.56 $\uparrow$ 0.30
	+INTCORT	79.31$\uparrow$9.37	83.69$\uparrow$7.30	72.97$\uparrow$0.61	82.95$\uparrow$1.70	94.42$\uparrow$3.27	99.40	100.00$\uparrow$2.74
Molmo 7B	Original model	28.77	61.37	61.18	63.92	50.77	69.28	73.78
	+AdaptVis	28.99 $\uparrow$ 0.22	62.23 $\uparrow$ 0.86	61.29 $\uparrow$ 0.11	64.49 $\uparrow$ 0.57	50.96 $\uparrow$ 0.19	69.88$\uparrow$0.60	73.78
	+DoLa	28.99 $\uparrow$ 0.22	63.09 $\uparrow$ 1.72	60.85 $\downarrow$ 0.33	64.77 $\uparrow$ 0.85	50.77	69.28	74.39 $\uparrow$ 0.61
	+VCD	30.71 $\uparrow$ 1.94	64.81 $\uparrow$ 3.44	58.06 $\downarrow$ 3.12	64.49 $\uparrow$ 0.57	51.15 $\uparrow$ 0.38	65.66 $\downarrow$ 3.62	80.79 $\uparrow$ 7.01
	+INTCORT	33.30$\uparrow$4.53	76.39$\uparrow$15.02	63.29$\uparrow$2.11	69.03$\uparrow$5.11	52.88$\uparrow$2.11	71.69$\uparrow$2.41	83.54$\uparrow$9.76
LLaVA-NeXT 7B	Original model	35.34	10.30	63.96	48.01	60.96	51.51	64.02
	+AdaptVis	48.06 $\uparrow$ 12.72	13.30 $\uparrow$ 3.00	64.63 $\uparrow$ 0.67	50.57$\uparrow$2.56	61.92 $\uparrow$ 0.96	97.29$\uparrow$45.78	77.74$\uparrow$13.72
	+DoLa	35.67 $\uparrow$ 0.33	8.15 $\downarrow$ 2.15	64.07 $\uparrow$ 0.11	47.73 $\downarrow$ 0.28	60.96	51.51	64.02
	+VCD	37.47 $\uparrow$ 2.13	14.74 $\uparrow$ 4.44	64.07 $\uparrow$ 0.11	48.58 $\uparrow$ 0.57	61.34 $\uparrow$ 0.38	63.13 $\uparrow$ 11.62	66.03 $\uparrow$ 2.01
	+INTCORT	48.92$\uparrow$13.58	17.60$\uparrow$7.30	65.24$\uparrow$1.28	49.43 $\uparrow$ 1.42	70.00$\uparrow$9.04	53.11 $\uparrow$ 1.60	64.94 $\uparrow$ 0.92

Table 2. Comparison with existing training-free inference methods. All values are accuracies (%).

Benchmark	LLaVA-NeXT 7B	+AdaptVis	+INTCORT	+AdaptVis +INTCORT
VG-one	35.34	48.06	48.92	62.02
VG-two	10.30	13.30	17.60	22.60
COCO-one	63.96	64.63	65.24	67.68
COCO-two	48.01	50.57	49.43	52.85
CV-Bench-R	60.96	61.92	70.00	72.77
Controlled-A	51.51	97.29	53.11	98.99
Controlled-B	64.02	77.74	64.94	84.01

Table 3. Complementarity analysis. All are accuracies (%).

Benchmark	INTCORT w/ Qwen2-0.5B- (Original) Instruct Router	w/ Mean Confidence	w/ Max Confidence
Controlled-A	99.10	98.09	93.37
VG-one	79.20	73.93	79.09

Table 4. Ablation results on Qwen2-VL-7B. All values are accuracies (%).

with AdaptVis achieves the best results, yielding average improvements of 11.47%, 24.8%, and 38.0% over AdaptVis alone, INTCORT alone and the original baseline, respectively. These results validate the strong complementarity between our method and AdaptVis.

3.4. Ablation

We conduct ablation studies on Qwen2-VL-7B using Controlled-A and VG-one. Specifically, we investigate three variants: re-

Benchmark	Qwen2-VL 7B	+INTCORT
POPE-P	87.77	87.77
POPE-R	88.93	88.93
MME	88.08	88.12

Table 5. Accuracy comparison on non-spatial benchmarks. All values are accuracies (%).

placing the original Qwen2.5-1.5B-Instruct spatial query classifier with Qwen2-0.5B-Instruct, and substituting min confidence aggregation with mean or max aggregation. The results in Table 4 show that all three modifications lead to performance drops, demonstrating the effectiveness of the INTCORT design.

4. CONCLUSION

This paper proposes INTCORT, a training-free framework for enhancing the spatial reasoning capability of VLMs. We show that incorrect spatial predictions from direct inference do not necessarily indicate the absence of spatial reasoning ability, as alternative inference views generated through input transformations can recover these failures. By aggregating multi-view predictions with relation-token confidence, INTCORT improves spatial reasoning without modifying the VLM’s internal mechanisms. Extensive experiments on six VLMs across seven spatial reasoning benchmarks demonstrate that INTCORT consistently improves performance over corresponding base models and existing training-free methods.

5. REFERENCES

- [1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al., “Flamingo: a visual language model for few-shot learning,” *Advances in neural information processing systems*, vol. 35, pp. 23716–23736, 2022.
- [2] Jingqi Xu, Jingxi Lu, Chenghao Li, Sreetama Sarkar, Souvik Kundu, and Peter A Beerel, “Redvtp: Training-free acceleration of diffusion vision-language models inference via masked token-guided visual token pruning,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 2783–2792.
- [3] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al., “Qwen technical report,” *arXiv preprint arXiv:2309.16609*, 2023.
- [4] Amita Kamath, Jack Hessel, and Kai-Wei Chang, “What’s “up” with vision-language models? investigating their struggle with spatial reasoning,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 9161–9175.
- [5] Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai C Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, et al., “Cambrian-1: A fully open, vision-centric exploration of multimodal llms,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 87310–87356, 2024.
- [6] Boyuan Chen, Zhuo Xu, Sean Kirmani, Brain Ichter, Dorsa Sadigh, Leonidas Guibas, and Fei Xia, “Spatialvlm: Endowing vision-language models with spatial reasoning capabilities,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 14455–14465.
- [7] Jingqi Xu, Jingxi Lu, Chenghao Li, Sreetama Sarkar, and Peter A Beerel, “Hivtp: A training-free method to improve vlms efficiency via hierarchical visual token pruning using middle-layer-based importance score,” *arXiv preprint arXiv:2509.23663*, 2025.
- [8] Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, and Manling Li, “Why is spatial reasoning hard for vlms? an attention mechanism perspective on focus areas,” *arXiv preprint arXiv:2503.01773*, 2025.
- [9] Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James R Glass, and Pengcheng He, “Dola: Decoding by contrasting layers improves factuality in large language models,” in *International Conference on Learning Representations*, 2024, vol. 2024, pp. 54158–54183.
- [10] Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing, “Mitigating object hallucinations in large vision-language models through visual contrastive decoding,” in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 13872–13882.
- [11] Yuexiang Zhai, Shengbang Tong, Xiao Li, Mu Cai, Qing Qu, Yong Jae Lee, and Yi Ma, “Investigating the catastrophic forgetting in multimodal large language models,” *arXiv preprint arXiv:2309.10313*, 2023.
- [12] Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A Shamma, et al., “Visual genome: Connecting language and vision using crowdsourced dense image annotations,” *International journal of computer vision*, vol. 123, no. 1, pp. 32–73, 2017.
- [13] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick, “Microsoft coco: Common objects in context,” in *European conference on computer vision*. Springer, 2014, pp. 740–755.
- [14] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al., “Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution,” *arXiv preprint arXiv:2409.12191*, 2024.
- [15] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin, “Qwen2.5-vl technical report,” 2025.
- [16] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, et al., “Qwen3-vl technical report,” *arXiv preprint arXiv:2511.21631*, 2025.
- [17] Matt Deitke, Christopher Clark, Sangho Lee, Rohun Tripathi, Yue Yang, Jae Sung Park, Mohammadreza Salehi, Niklas Muennighoff, Kyle Lo, Luca Soldaini, et al., “Molmo and pixmo: Open weights and open data for state-of-the-art vision-language models,” in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 91–104.
- [18] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee, “Llavanext: Improved reasoning, ocr, and world knowledge,” 2024.
- [19] Hongxing Li, Dingming Li, Zixuan Wang, Yuchen Yan, Hang Wu, Wenqi Zhang, Yongliang Shen, Weiming Lu, Jun Xiao, and Yueting Zhuang, “Spatalladder: Progressive training for spatial reasoning in vision-language models,” in *International Conference on Learning Representations*, 2026, vol. 2026, pp. 76566–76592.
- [20] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Xin Zhao, and Ji-Rong Wen, “Evaluating object hallucination in large vision-language models,” in *Proceedings of the 2023 conference on empirical methods in natural language processing*, 2023, pp. 292–305.
- [21] Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, et al., “Mme: A comprehensive evaluation benchmark for multimodal large language models,” *Advances in Neural Information Processing Systems*, vol. 38, 2026.